

Problem Set 3: Membrane Transport

Due: Sunday, September 27 at 11:59 PM Value: 25 points Format: Five-question problem set

Purpose

Predict movement across membranes using concentration gradients, permeability, electrochemical forces, and tonicity.

Your task

- Q1 (5): A membrane separates 100 mM glucose from 20 mM glucose and is permeable only to glucose. Predict net movement and the eventual equilibrium condition.
- Q2 (5): Explain why water can move toward a solution with a higher concentration of a nonpenetrating solute even though the solute does not cross.
- Q3 (5): An animal cell is placed in 300 mM urea. Urea crosses the membrane slowly; intracellular nonpenetrating solute is 300 mOsm. Predict cell volume immediately and after equilibration.
- Q4 (5): Compare facilitated diffusion with primary active transport in terms of energy source, direction, saturation, and specificity.
- Q5 (5): Use the channel-current data below to decide whether opening probability or single-channel conductance changes after treatment.

Provided evidence or data

Condition	Open probability	Current per open channel
Control	0.20	2.1 pA
Treatment	0.62	2.0 pA

Submit

- One PDF with diagrams where helpful and complete reasoning.

Grading criteria

Criterion	Pts	Full-credit evidence
Question 1	5	Correct setup, units, and biological interpretation.
Question 2	5	Correct reasoning shown; not just a final answer.
Question 3	5	Mechanism-based explanation uses appropriate evidence.
Question 4	5	Prediction follows from the stated model and conditions.
Question 5	5	Data are represented or evaluated accurately.

Submission standards

Upload a readable PDF unless the handout specifies another format. Include your name, course code, and assignment title. Cite borrowed ideas and retain the editable source file. The learning portal timestamp is the official submission time.